For each image, let P_0 be the initial binary prediction, P_j its prediction under shifted prompt j, and G the manual reference mask. For binary masks U, V with at least one foreground pixel between them, Dice agreement is

$$D(U,V)=\frac{2|U\cap V|}{|U|+|V|}. \tag{1}$$

Here $|\cdot|$ counts foreground pixels. For consistency scoring, Dice is defined as one when both predicted masks are empty. Method A uses the confidence risk in Eq. (2):

$$r_{\mathrm{A}}=1-\hat{q}. \tag{2}$$

Here $\hat{q}$ denotes the raw predicted intersection over union (IoU) returned by the SAM quality head for the initial mask. For method B, we shifted the initial clipped box left and right by 5 % of its width and up and down by 5 % of its height, clipping each shifted box to the image. The four initial-to-shifted Dice values define the disagreement risk

$$r_{\mathrm{B}}=1-\frac{1}{4}\sum_{j=1}^{4} D(P_0,P_j). \tag{3}$$

All four directions were retained when clipping produced coincident boxes. These deterministic shifts probe local stability relative to box size. Method B adds four mask-decoder passes while reusing the image embedding. Recorded inference times cover the combined pipeline; A-only and incremental B times were not separately benchmarked.

The quality reference for both methods was the initial prediction's Dice against the manual mask, $D(P_0,G)$ in Eq. (1). An empty initial prediction against the nonempty reference has Dice zero. Low quality was defined as reference Dice below 0.70, an operational cutoff for this evaluation. The raw model confidence in Eq. (2) and prediction-consistency score in Eq. (3) remain separate from this reference-quality outcome.

OpenAI Codex assisted with study-design discussions and implementation of research, statistical and plotting code. The pipeline processed public images with frozen SAM weights, and figures were rendered conventionally from recorded data and captured predictions. The human authors retain responsibility for the implementation and calculated outputs. A precise underlying Codex model identifier was not available in the archived study records.

Development threshold selection

For each method, we searched the distinct development risk scores and the option to flag zero cases. A case was flagged if its score met or exceeded the candidate threshold. The selected

Table S12. Post hoc diagnostic-stratum symmetric Kitagawa decomposition of external-minus-internal differences.

Method: outcome	Overall difference, pp [95 % interval]	Composition-associated term, pp [95 % interval]	Within-stratum term, pp [95 % interval]	Valid overall n	Valid component n
A: Detection	-13.84 [-40.85, +11.40]	-1.62 [-10.89, +5.73]	-12.23 [-39.94, +14.30]	5000	4999
A: Review fraction	-6.56 [-11.54, -1.31]	+0.64 [-0.01, +1.54]	-7.20 [-12.15, -2.08]	5000	5000
B: Detection	+8.13 [-15.09, +28.18]	-0.78 [-6.43, +3.62]	+8.91 [-14.32, +28.70]	5000	4999
B: Review fraction	-4.53 [-9.31, +0.41]	+0.79 [-0.01, +1.80]	-5.32 [-10.03, -0.56]	5000	5000

Post hoc descriptive accounting at the locked 20 %-development-budget thresholds (A=0.13705134391784668; B=0.06972856653991777), without recalibration. Direction is external minus internal. Strata are dataset-reported benign and malignant diagnoses; not every diagnosis was histopathologically confirmed.

For diagnostic stratum s, let $w_{c,s}$ be its weight and $p_{c,s}$ its outcome rate in cohort c, with E denoting external and I internal testing. For detection, weights are diagnostic shares among low-quality patients and rates are stratum-specific low-quality detection. For review, weights are diagnostic shares among all scored patients and rates are stratum-specific review fractions. Rates in Eqs. (S1) and (S2) are fractions; Table S12 reports differences in percentage points. The symmetric Kitagawa composition-associated term and within-stratum-rate term are

$$C = \sum_s \left(w_{\mathrm{E},s} - w_{\mathrm{I},s}\right) \frac{p_{\mathrm{E},s} + p_{\mathrm{I},s}}{2}, \tag{S1}$$

$$W = \sum_s \left(p_{\mathrm{E},s} - p_{\mathrm{I},s}\right) \frac{w_{\mathrm{E},s} + w_{\mathrm{I},s}}{2}. \tag{S2}$$

Together, C and W sum to the external-minus-internal overall rate difference when the stratum rates are defined. Weights and rates are recomputed in each bootstrap draw.

Intervals are nominal, unadjusted descriptive percentile 95 % intervals from 5,000 independent within-cohort patient bootstrap replicates (seed 20261107; NumPy PCG64; linear finite-value quantiles); A/B, quality and diagnosis remain paired. Duplicate draws are repeated observations of existing patients, not new cases.

Overall detection differences have 5,000 valid replicates. One draw has no external benign low-quality case, making its detection stratum rate undefined; the NA propagates to both detection components, leaving 4,999 valid component replicates. Review-fraction components and overall differences have 5,000 valid replicates. Undefined rates are not imputed even when their stratum weight is zero.

Two diagnostic strata cannot remove device, acquisition, annotation or other confounding. This decomposition is not a causal attribution and no percent-explained claim is made. Percentile intervals supplement the original H1–H3 Newcombe intervals; crossing zero does not demonstrate equivalence.

Direct standardization uses the arithmetic mean of the two cohort weights for each diagnostic stratum. When the stratum rates are defined, the resulting standardized external-minus-internal difference equals the within-stratum-rate term W in Eq. (S2). Overall rates are calculated directly from aggregate numerators and denominators. Components may not sum to the displayed total after rounding. Method: Kitagawa EM, J Am Stat Assoc 1955;50:1168–1194, doi:10.1080/01621459.1955.10501299.